

Production and Costs

**ECON 101: Introduction to
Microeconomics**

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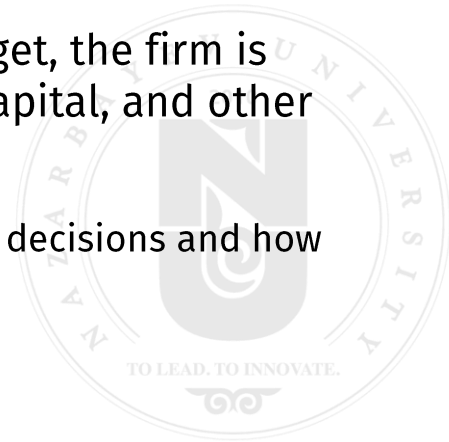


The Production Decisions of a Firm

The production decisions of firms can be understood in three steps:

1. **Production Technology:** How *inputs* (such as labor, capital, and raw materials) can be transformed into *outputs*.
2. **Cost Constraints:** How *inputs* (such as labor, capital, and raw materials) can be transformed into *outputs*.
3. **Input Choices:** Just as a consumer is constrained by a limited budget, the firm is concerned about the cost of production and the *prices* of labor, capital, and other inputs.

theory of the firm Explanation of how a firm makes cost-minimizing production decisions and how its cost varies with its output.



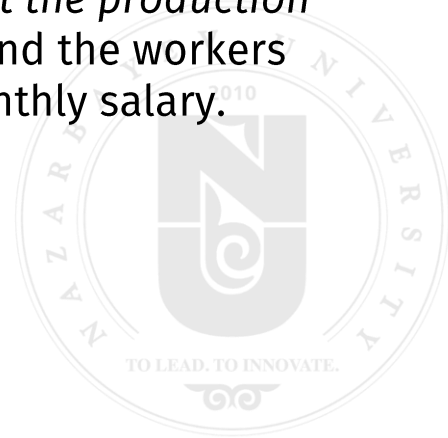
Firms and Their Production Decisions

Why Do Firms Exist?

Firms offer a means of *coordination* that is extremely important and would be sorely missing if workers operated independently.

Firms eliminate the need for every worker to negotiate every task that he or she will perform, and bargain over the fees that will be paid for those tasks.

Firms can avoid this kind of bargaining by having managers that *direct the production of salaried workers*—they tell workers what to do and when to do it, and the workers (as well as the managers themselves) are simply paid a weekly or monthly salary.



Firms and Their Production Decisions

The Technology of Production

factors of production Inputs into the production process (e.g., labor, capital, and materials).

We can divide inputs into the broad categories of *labor*, *materials* and *capital*, each of which might include more narrow subdivisions.

Labor inputs include skilled workers (carpenters, engineers) and unskilled workers (agricultural workers), as well as the entrepreneurial efforts of the firm's managers. Materials include steel, plastics, electricity, water, and any other goods that the firm buys and transforms into final products.

Capital includes land, buildings, machinery and other equipment, as well as inventories.



Firms and Their Production Decisions

The Production Function

production function Function showing the highest output that a firm can produce for every specified combination of inputs.

$$q = F(K, L)$$

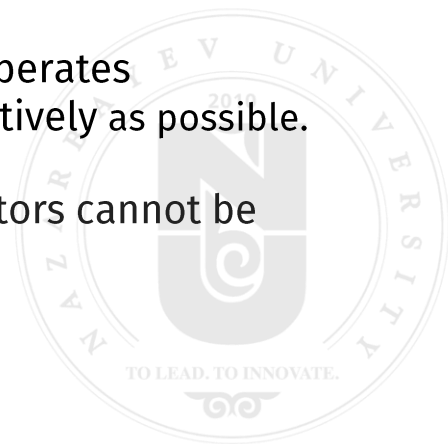
Production functions describe what is *technically feasible* when the firm operates *efficiently*—that is, when the firm uses each combination of inputs as effectively as possible.

The Short Run versus the Long Run

short run Period of time in which quantities of one or more production factors cannot be changed.

fixed input Production factor that cannot be varied.

long run Amount of time needed to make all production inputs variable



Production with one variable input.

Average and Marginal Products

average product Output per unit of a particular input.

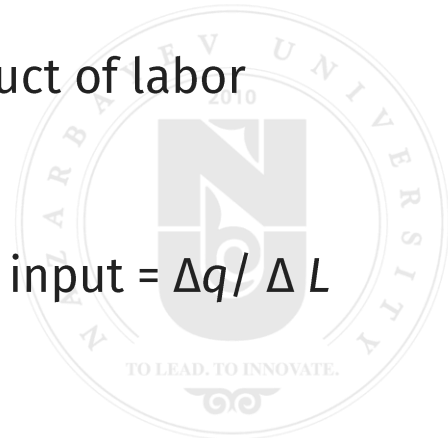
marginal product Additional output produced as an input is increased by one unit.

Remember that the marginal product of labor depends on the amount of capital used.

If the capital input increased from 10 to 20, the marginal product of labor most likely would increase.

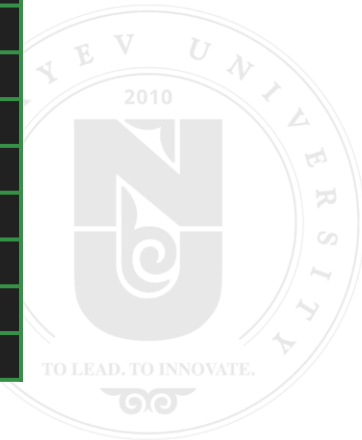
Average product of labor = Output/labor input = q/L

Marginal product of labor = Change in output/change in labor input = $\Delta q / \Delta L$



Production with one variable input

AMOUNT OF LABOR (L)	AMOUNT OF CAPITAL (K)	TOTAL OUTPUT (q)	AVERAGE PRODUCT (q/L)	MARGINAL PRODUCT ($\Delta q/\Delta L$)
0	10	0	—	—
1	10	15	15	15
2	10	40	20	25
3	10	69	23	29
4	10	96	24	27
5	10	120	24	24
6	10	138	23	18
7	10	147	21	9
8	10	152	19	5
9	10	153	17	1
10	10	150	15	-3
11	10	143	13	-7
12	10	133	11.08	-10



The Slopes of the Product Curve

PRODUCTION WITH ONE VARIABLE INPUT

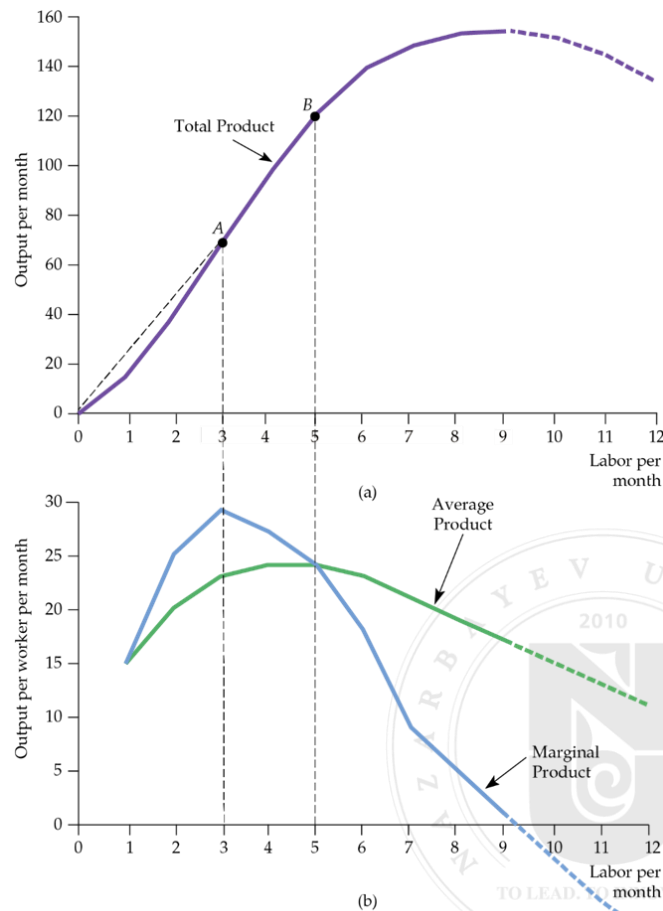
The total output curve in (a) shows the output produced for different amounts of labor input.

The average and marginal products in (b) can be obtained (using the data in Table 6.1) from the total product curve.

At point A in (a), with 3 units of labor, the marginal product is 29 because the tangent to the total product curve has a slope of 29.

The average product of labor, however, is 23, which is the slope of the line from the origin to point A. Also, the marginal product of labor reaches its maximum at this point.

At point B, with 5 units of labor, the marginal product of labor has dropped to 24 and is equal to the average product of labor.



The Slopes of the Product Curve

PRODUCTION WITH ONE VARIABLE INPUT

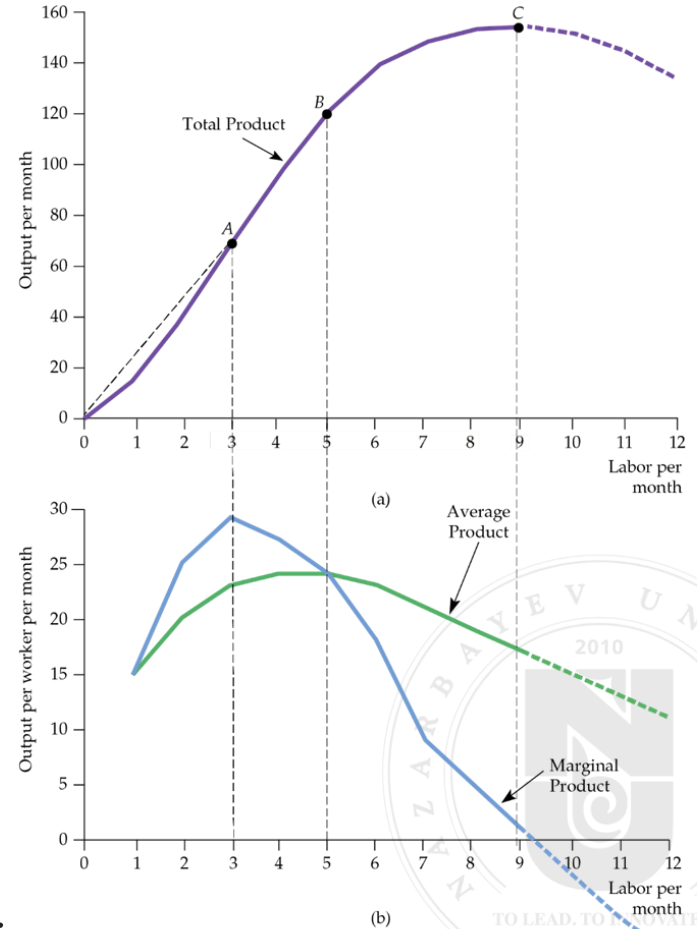
Thus, in (b), the average and marginal product curves intersect (at point D).

Note that when the marginal product curve is above the average product, the average product is increasing.

When the labor input is greater than 5 units, the marginal product is below the average product, so the average product is falling. Once the labor input exceeds 9 units, the marginal product becomes negative, so that total output falls as more labor is added.

At C, when total output is maximized, the slope of the tangent to the total product curve is 0, as is the marginal product.

Beyond that point, the marginal product becomes negative.



The Average Product of Labor Curve

In general, the average product of labor is given by the slope of the line drawn from the origin to the corresponding point on the total product curve.

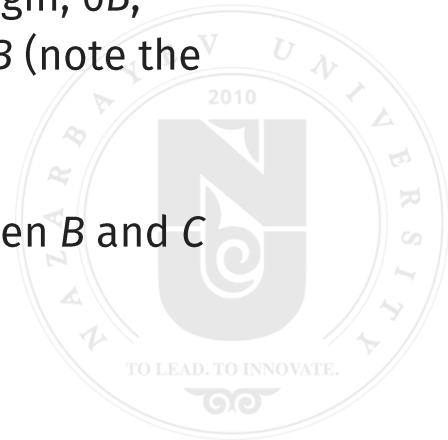
The Marginal Product of Labor Curve

In general, the marginal product of labor at a point is given by the slope of the total product at that point.

THE RELATIONSHIP BETWEEN THE AVERAGE AND MARGINAL PRODUCTS

In the figures, the average product is the slope of the line from the origin, OB , and the marginal product is the tangent to the total product curve at B (note the equality of the average and marginal products at point D).

The slope of the tangent to the total product curve at any point between B and C is lower than the slope of the line from the origin.



The Law of Diminishing Marginal Returns

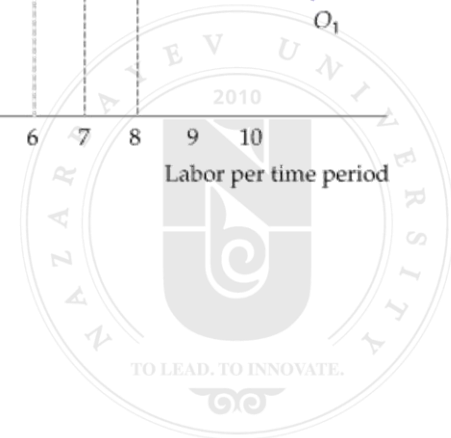
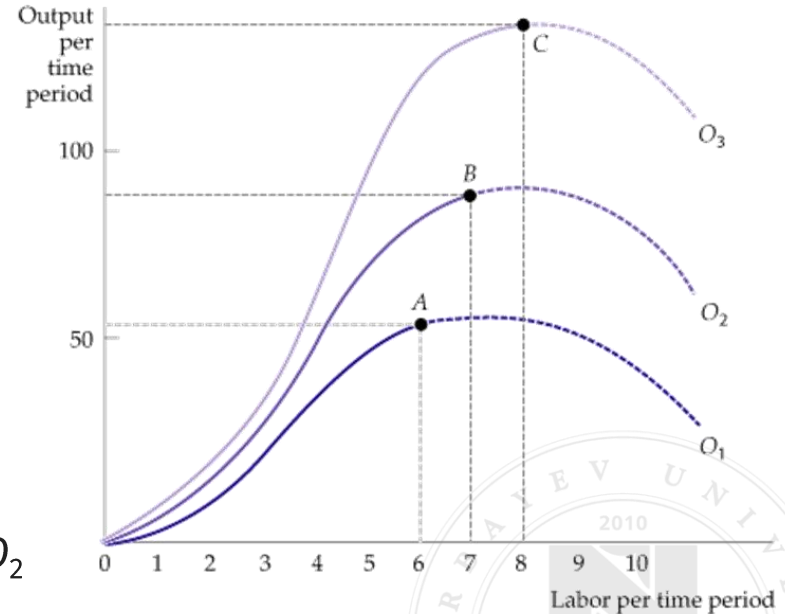
law of diminishing marginal returns

Principle that as the use of an input increases with other inputs fixed, the resulting additions to output will eventually decrease.

THE EFFECT OF TECHNOLOGICAL IMPROVEMENT

Labor productivity (output per unit of labor) can increase if there are improvements in technology, even though any given production process exhibits diminishing returns to labor.

As we move from point A on curve O_1 to B on curve O_2 to C on curve O_3 over time, labor productivity increases.



Production with Two Variable Inputs

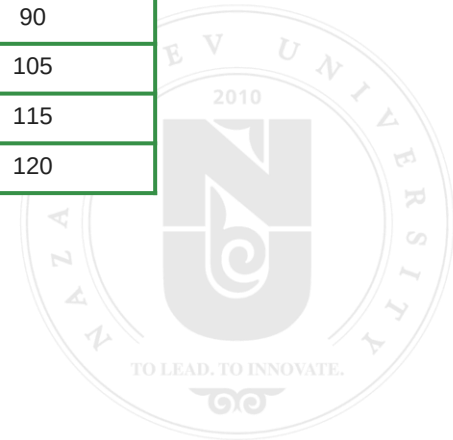
Isoquants

Isoquants Curve showing all possible combinations of inputs that yield the same output.

CAPITAL INPUT	LABOR INPUT				
	1	2	3	4	5
1	20	40	55	65	75
2	40	60	75	85	90
3	55	75	90	100	105
4	65	85	100	110	115
5	75	90	105	115	120

ISOQUANT MAPS

isoquant map Graph combining a number of isoquants, used to describe a production function.

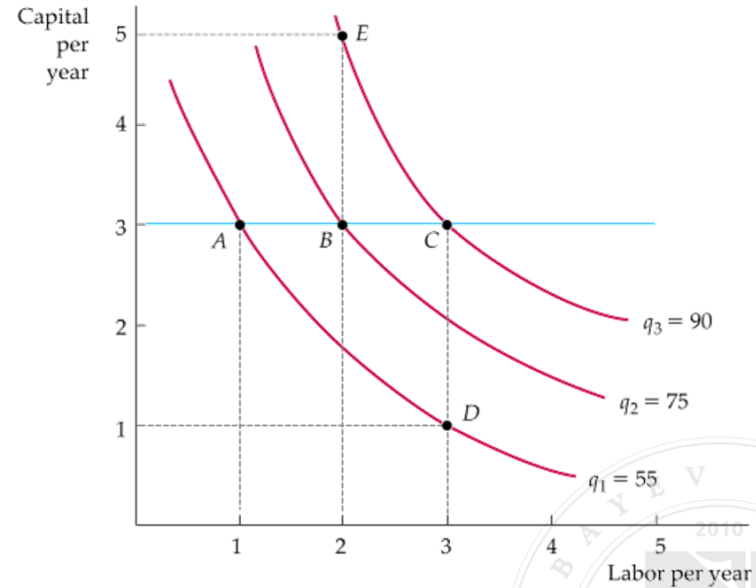


Production with Two Variable Inputs

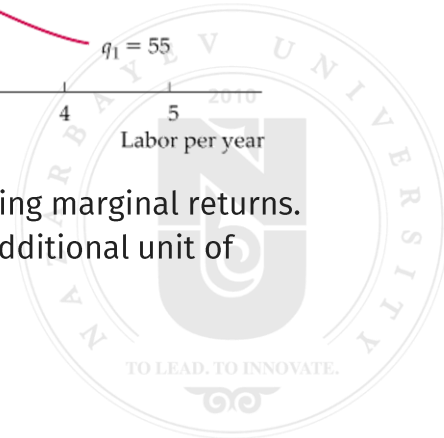
PRODUCTION WITH TWO VARIABLE INPUTS

A set of isoquants, or isoquant map, describes the firm's production function.

Output increases as we move from isoquant q_1 (at which 55 units per year are produced at points such as A and D), to isoquant q_2 (75 units per year at points such as B), and to isoquant q_3 (90 units per year at points such as C and E).



By drawing a horizontal line at a particular level of capital—say 3, we can observe diminishing marginal returns. Reading the levels of output from each isoquant as labor is increased, we note that each additional unit of labor generates less and less additional output.



Production with Two Variable Inputs

Input Flexibility

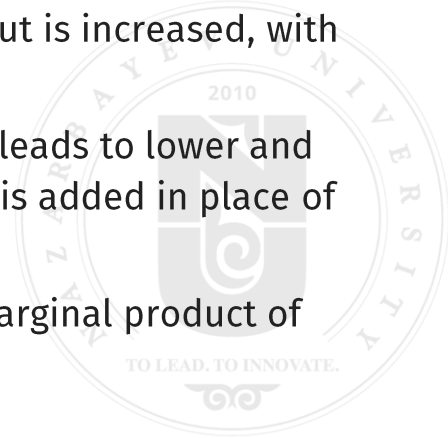
Isoquants show the flexibility that firms have when making production decisions: They can usually obtain a particular output by substituting one input for another. It is important for managers to understand the nature of this flexibility.

Diminishing Marginal Returns

Even though both labor and capital are variable in the long run, it is useful for a firm that is choosing the optimal mix of inputs to ask what happens to output as each input is increased, with the other input held fixed.

Because adding one factor while holding the other factor constant eventually leads to lower and lower incremental output, the isoquant must become steeper as more capital is added in place of labor and flatter when labor is added in place of capital.

There are also diminishing marginal returns to capital. With labor fixed, the marginal product of capital decreases as capital is increased.



Production with Two Variable Inputs

Substitution Among Inputs

marginal rate of technical substitution (MRTS) Amount by which the quantity of one input can be reduced when one extra unit of another input is used, so that output remains constant.

$$\begin{aligned}\text{MRTS} &= -\text{Change in capital input/change in labor input} \\ &= -\Delta K/\Delta L \text{ (for a fixed level of } q\text{)}\end{aligned}$$

DIMINISHING MRTS

Additional output from increased use of labor = $(MP_L)(\Delta L)$

Reduction in output from decreased use of capital = $(MP_K)(\Delta K)$

Reduction in output from decreased use of capital = $(MP_K)(\Delta K)$

$$(MPL) (\Delta L) + (MPK)(\Delta K) = 0$$

Now, by rearranging terms we see that $(MPL)/(MPK) = -(\Delta K/\Delta L) = \text{MRTS}$

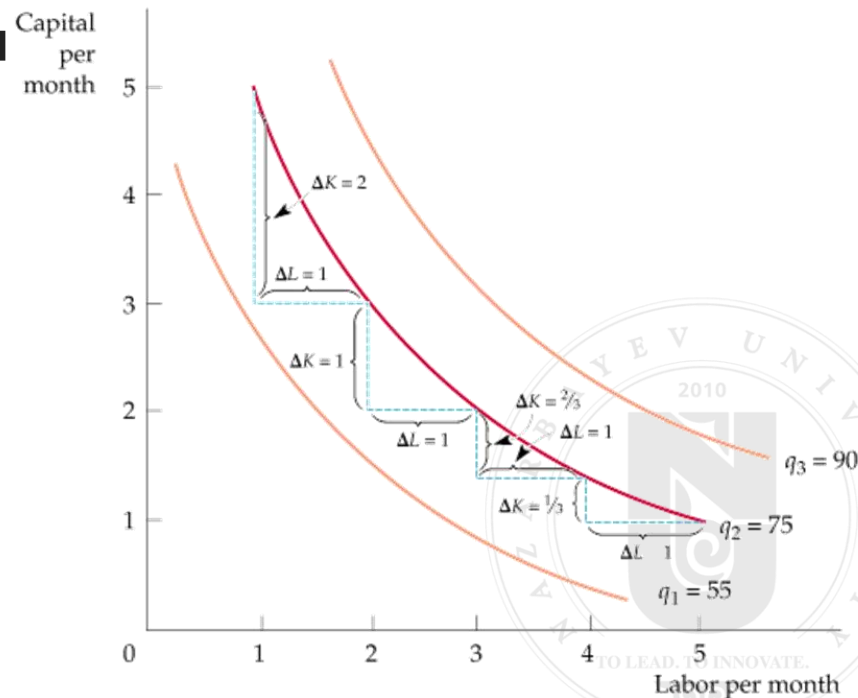


Production with Two Variable Inputs

MARGINAL RATE OF TECHNICAL SUBSTITUTION

Like indifference curves, isoquants are downward sloping and convex. The slope of the isoquant at any point measures the marginal rate of technical substitution—the ability of the firm to replace capital with labor while maintaining the same level of output.

On isoquant q_2 , the MRTS falls from 2 to 1 to $2/3$ to $1/3$.



Production with Two Variable Inputs

Production Functions—Two Special Cases

Two extreme cases of production functions show the possible range of input substitution in the production process: the case of ***perfect substitutes*** and the **fixed proportions production function**, sometimes called a *Leontief production function*.

fixed-proportions production function Production function with L-shaped isoquants, so that only one combination of labor and capital can be used to produce each level of output.

The fixed-proportions production function describes situations in which methods of production are limited.

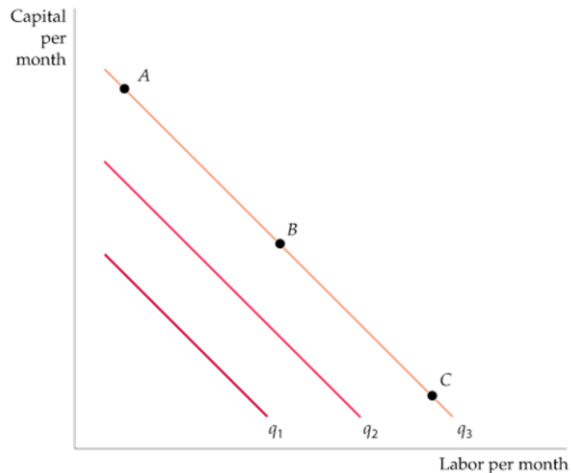


Production with Two Variable Inputs

ISOQUANTS WHEN INPUTS ARE PERFECT SUBSTITUTES

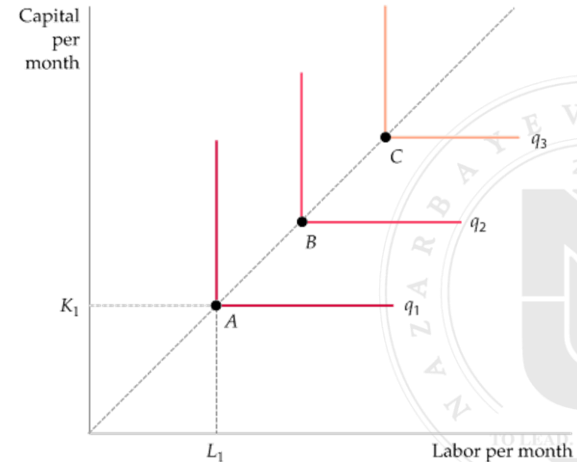
When the isoquants are straight lines, the MRTS is constant. Thus the rate at which capital and labor can be substituted for each other is the same no matter what level of inputs is being used.

Points A, B, and C represent three different capital-labor combinations that generate the same output q_3 .



FIXED-PROPORTIONS PRODUCTION FUNCTION

When the isoquants are L-shaped, only one combination of labor and capital can be used to produce a given output (as at point A on isoquant q_1 , point B on isoquant q_2 , and point C on isoquant q_3). Adding more labor alone does not increase output, nor does adding more capital alone.



Returns to Scale

returns to scale Rate at which output increases as inputs are increased proportionately.

INCREASING RETURNS TO SCALE

increasing returns to scale Situation in which output more than doubles when all inputs are doubled.

CONSTANT RETURNS TO SCALE

constant returns to scale Situation in which output doubles when all inputs are doubled.

DECREASING RETURNS TO SCALE

decreasing returns to scale Situation in which output less than doubles when all inputs are doubled.



Returns to Scale

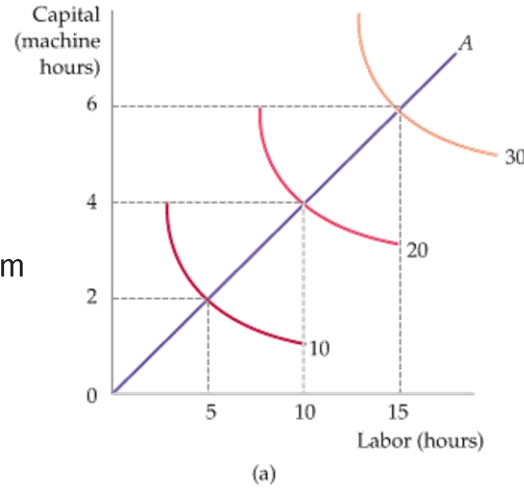
Describing Returns to Scale

Returns to scale need not be uniform across all possible levels of output. For example, at lower levels of output, the firm could have increasing returns to scale, but constant and eventually decreasing returns at higher levels of output.

In Figure (a), the firm's production function exhibits constant returns. Twice as much of both inputs is needed to produce 20 units, and three times as much is needed to produce 30 units.

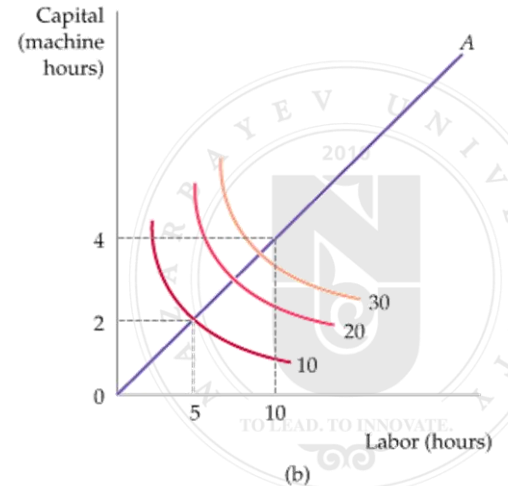
In Figure (b), the firm's production function exhibits increasing returns to scale. *Less* than twice the amount of both inputs is needed to increase production from 10 units to 20; substantially less than three times the inputs are needed to produce 30 units.

Returns to scale vary considerably across firms and industries. Other things being equal, the greater the returns to scale, the larger the firms in an industry are likely to be.



When a firm's production process exhibits constant returns to scale as shown by a movement along line OA in part (a), the isoquants are equally spaced as output increases proportionally.

However, when there are increasing returns to scale as shown in (b), the isoquants move closer together as inputs are increased along the line.



The Cost of Production

Which costs matter?

Economic Cost versus Accounting Cost

accounting cost Actual expenses plus depreciation charges for capital equipment.

economic cost Cost to a firm of utilizing economic resources in production.

Opportunity Cost

opportunity cost Cost associated with opportunities forgone when a firm's resources are not put to their best alternative use.

The concept of opportunity cost is particularly useful in situations where alternatives that are forgone do not reflect monetary outlays.

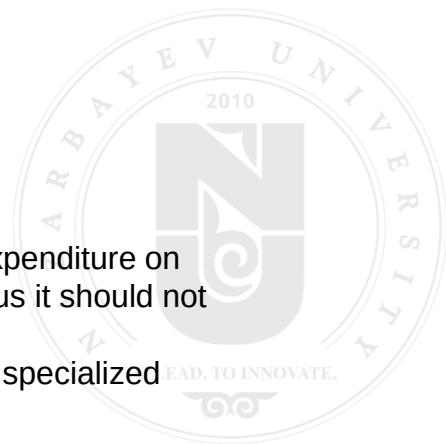
$$\text{Economic cost} = \text{Opportunity cost}$$

sunk cost Expenditure that has been made and cannot be recovered.

Because a sunk cost cannot be recovered, it should not influence the firm's decisions.

For example, if specialized equipment for a plant cannot be converted for alternative use, the expenditure on this equipment is a sunk cost. *Because it has no alternative use, its opportunity cost is zero.* Thus it should not be included as part of the firm's economic costs.

A *prospective* sunk cost is an *investment*. Here the firm must decide whether that investment in specialized equipment is *economical*.



Measuring Cost: Which Costs Matter?

● Fixed Costs and Variable Costs

total cost (TC or C) Total economic cost of production, consisting of fixed and variable costs.

fixed cost (FC) Cost that does not vary with the level of output and that can be eliminated only by shutting down.

variable cost (VC) Cost that varies as output varies.

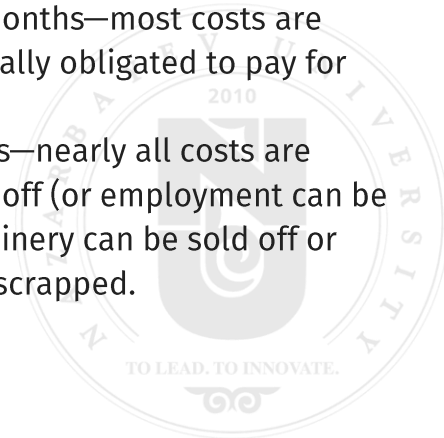
Fixed cost does not vary with the level of output—it must be paid even if there is no output. *The only way that a firm can eliminate its fixed costs is by shutting down.*

SHUTTING DOWN

Shutting down doesn't necessarily mean going out of business. By reducing the output of that factory to zero, the company could eliminate the costs of raw materials and much of the labor, but it would still incur the fixed costs of paying the factory's managers, security guards, and ongoing maintenance. The only way to eliminate those fixed costs would be to close the doors, turn off the electricity, and perhaps even sell off or scrap the machinery.

FIXED OR VARIABLE?

How do we know which costs are fixed and which are variable? Over a very short time horizon—say, a few months—most costs are fixed. Over such a short period, a firm is usually obligated to pay for contracted shipments of materials. Over a very long time horizon—say, ten years—nearly all costs are variable. Workers and managers can be laid off (or employment can be reduced by attrition), and much of the machinery can be sold off or not replaced as it becomes obsolete and is scrapped.



Amortizing Sunk Costs

amortization Policy of treating a one-time expenditure as an annual cost spread out over some number of years.

Amortizing large capital expenditures and treating them as ongoing fixed costs can simplify the economic analysis of a firm's operation. As we will see, treating capital expenditures this way can make it easier to understand the tradeoff that a firm faces in its use of labor versus capital.

When distinguishing sunk from fixed costs does become essential to the economic analysis, we will let you know.



SUNK, FIXED, AND VARIABLE COSTS

It is important to understand the characteristics of production costs and to be able to identify which costs are fixed, which are variable, and which are sunk.

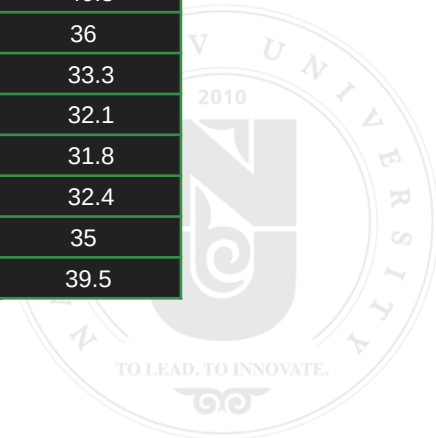
Computers: Because computers are very similar, competition is intense, and profitability depends on the ability to keep costs down. Most important are the cost of components and labor.

Software: A software firm will spend a large amount of money to develop a new application. The company can recoup its investment by selling as many copies of the program as possible.

Pizzas: For the pizzeria, sunk costs are fairly low because equipment can be resold if the pizzeria goes out of business. Variable costs are low—mainly the ingredients for pizza and perhaps wages for a workers to produce and deliver pizzas.

Measuring Cost: Which Costs Matter?

RATE OF OUTPUT (UNITS PER YEAR)	FIXED COST (DOLLARS PER YEAR)	VARIABLE COST (DOLLARS PER YEAR)	TOTAL COST (DOLLARS PER YEAR)	MARGINAL COST (DOLLARS PER UNIT)	AVERAGE FIXED COST (DOLLARS PER UNIT)	AVERAGE VARIABLE COST (DOLLARS PER UNIT)	AVERAGE TOTAL COST (DOLLARS PER UNIT)
	(FC) (1)	(VC) (2)	(TC) (3)	(MC) (4)	(AFC) (5)	(AVC) (6)	(ATC) (7)
0	50	0	50	—	—	—	—
1	50	50	100	50	50	50	100
2	50	78	128	28	25	39	64
3	50	98	148	20	16.7	32.7	49.3
4	50	112	162	14	12.5	28	40.5
5	50	130	180	18	10	26	36
6	50	150	200	20	8.3	25	33.3
7	50	175	225	25	7.1	25	32.1
8	50	204	254	29	6.3	25.5	31.8
9	50	242	292	38	5.6	26.9	32.4
10	50	300	350	58	5	30	35
11	50	385	435	85	4.5	35	39.5



Measuring Cost: Which Costs Matter?

MARGINAL COST (MC)

marginal cost (MC) Increase in cost resulting from the production of one extra unit of output.

Because fixed cost does not change as the firm's level of output changes, marginal cost is equal to the increase in variable cost or the increase in total cost that results from an extra unit of output. We can therefore write marginal cost as

$$MC = \Delta VC / \Delta q = \Delta TC / \Delta q$$

AVERAGE TOTAL COST (ATC)

average total cost (ATC) Firm's total cost divided by its level of output.

average fixed cost (AFC) Fixed cost divided by the level of output.

average variable cost (AVC) Variable cost divided by the level of output.



Measuring Cost: Which Costs Matter?

- **The Determinants of Short-Run Cost**

The change in variable cost is the per-unit cost of the extra labor w times the amount of extra labor needed to produce the extra output ΔL . Because $\Delta VC = w\Delta L$, it follows that

$$MC = \Delta VC / \Delta q = w\Delta L / \Delta q$$

The extra labor needed to obtain an extra unit of output is $\Delta L / \Delta q = 1/MP_L$. As a result,

$$MC = w / MP_L$$

DIMINISHING MARGINAL RETURNS AND MARGINAL COST

Diminishing marginal returns means that the marginal product of labor declines as the quantity of labor employed increases.

As a result, when there are diminishing marginal returns, marginal cost will increase as output increases.



The Shapes of the Cost Curves

COST CURVES FOR A FIRM

In **(a)** total cost TC is the vertical sum of fixed cost FC and variable cost VC.

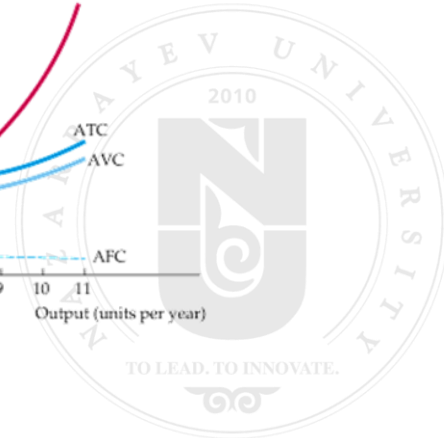
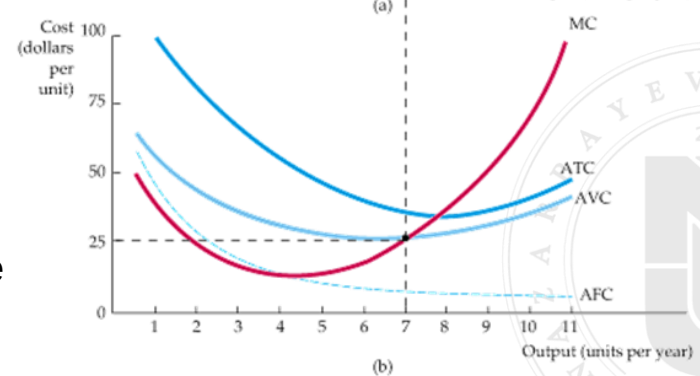
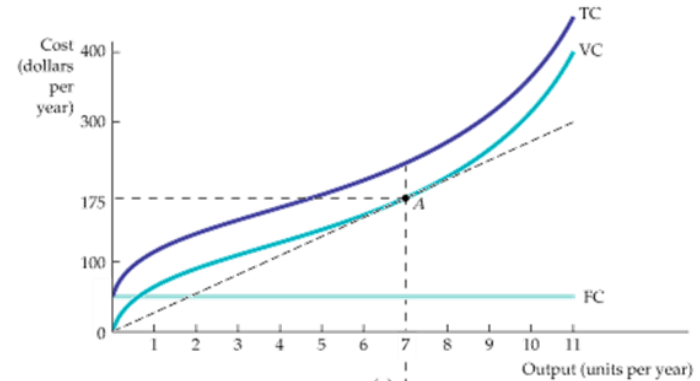
In **(b)** average total cost ATC is the sum of average variable cost AVC and average fixed cost AFC.

Marginal cost MC crosses the average variable cost and average total cost curves at their minimum points.

THE AVERAGE-MARGINAL RELATIONSHIP

Marginal and average costs are another example of the average-marginal relationship described in Chapter 6 (with respect to marginal and average product).

Because average total cost is the sum of average variable cost and average fixed cost and the AFC curve declines everywhere, the vertical distance between the ATC and AVC curves decreases as output increases.



Example: THE SHORT-RUN COST OF ALUMINUM SMELTING

The production of aluminum begins with the mining of bauxite. The process used to separate the oxygen atoms from aluminum oxide molecules, called smelting, is the most costly step in producing aluminum.

The expenditure on a smelting plant, although substantial, is a sunk cost and can be ignored. Fixed costs are relatively small and can also be ignored.

PRODUCTION COSTS FOR ALUMINUM SMELTING (\$/TON) (BASED ON AN OUTPUT OF 600 TONS/DAY)

PER-TON COSTS THAT ARE CONSTANT FOR ALL OUTPUT LEVELS	OUTPUT $\leq$ 600 TONS/DAY	OUTPUT $\geq$ 600 TONS/DAY
Electricity	\$316	\$316
Alumina	369	369
Other raw materials	125	125
Plant power and fuel	10	10
Subtotal	\$820	\$820
PER-TON COSTS THAT INCREASE WHEN OUTPUT EXCEEDS 600 TONS/DAY		
Labor	\$150	\$225
Maintenance	120	180
Freight	50	75
Subtotal	\$320	\$480
Total per-ton production costs	\$1140	\$1300



THE SHORT-RUN COST OF ALUMINUM SMELTING

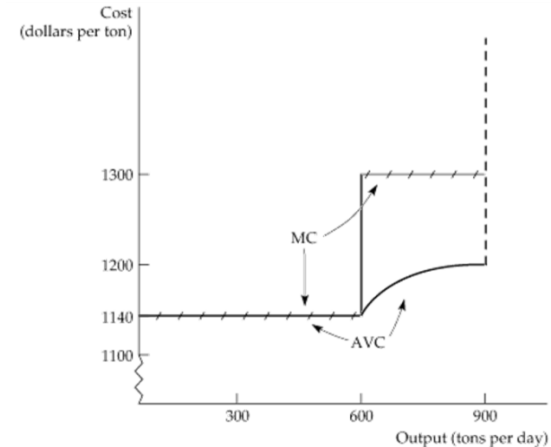
For an output q up to 600 tons per day, total variable cost is $\$1140q$, so marginal cost and average variable cost are constant at $\$1140$ per ton. If we increase production beyond 600 tons per day by means of a third shift, the marginal cost of labor, maintenance, and freight increases from $\$320$ per ton to $\$480$ per ton, which causes marginal cost as a whole to increase from $\$1140$ per ton to $\$1300$ per ton. What happens to average variable cost when output is greater than 600 tons per day? When $q > 600$, total variable cost is given by:

$$TVC = (1140)(600) + 1300(q - 600) = 1300q - 96,000$$

Therefore average variable cost is

$$AVC = 1300 - \frac{96,000}{q}$$

The short-run average variable cost of smelting is constant for output levels using up to two labor shifts. When a third shift is added, marginal cost and average variable cost increase until maximum capacity is reached.



Cost in the Long Run

● The User Cost of Capital

user cost of capital Annual cost of owning and using a capital asset, equal to economic depreciation plus forgone interest.

The user cost of capital is given by the *sum of the economic depreciation and the interest (i.e., the financial return) that could have been earned had the money been invested elsewhere*. Formally,

$$\text{User Cost of Capital} = \text{Economic Depreciation} + (\text{InterestRate})(\text{Value of Capital})$$

We can also express the user cost of capital as a *rate* per dollar of capital:

$$r = \text{Depreciation rate} + \text{Interest rate}$$



Cost in the Long Run

The Cost-Minimizing Input Choice

We now turn to a fundamental problem that all firms face: *how to select inputs to produce a given output at minimum cost.*

For simplicity, we will work with two variable inputs: labor (measured in hours of work per year) and capital (measured in hours of use of machinery per year).

THE PRICE OF CAPITAL

In the long run, the firm can adjust the amount of capital it uses. The firm must decide *prospectively* how much capital to obtain.

In order to compare the firm's expenditure on capital with its ongoing cost of labor, we want to express this capital expenditure as a *flow*—*e.g.*, in dollars per year. To do this, we must amortize the expenditure by spreading it over the lifetime of the capital, and we must also account for the forgone interest that the firm could have earned by investing the money elsewhere.

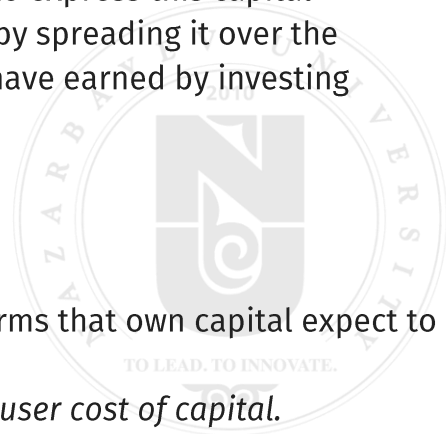
The price of capital is its user cost, given by $r = \text{Depreciation rate} + \text{Interest rate}$.

THE RENTAL RATE OF CAPITAL

rental rate Cost per year of renting one unit of capital.

If the capital market is competitive, *the rental rate should be equal to the user cost, r .* Why? Firms that own capital expect to earn a competitive return when they rent it. *This competitive return is the user cost of capital.*

Capital that is purchased can be treated as though it were rented at a rental rate equal to the user cost of capital.



Cost in the Long Run

- **The Isocost Line**

isocost line Graph showing all possible combinations of labor and capital that can be purchased for a given total cost.

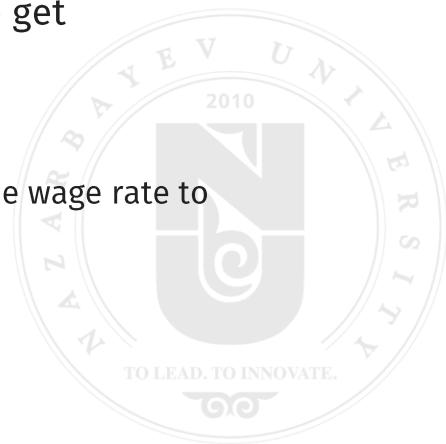
To see what an isocost line looks like, recall that the total cost C of producing any particular output is given by the sum of the firm's labor cost wL and its capital cost rK :

$$C = wL + rK$$

If we rewrite the total cost equation as an equation for a straight line, we get

$$K = C/r - (w/r)L$$

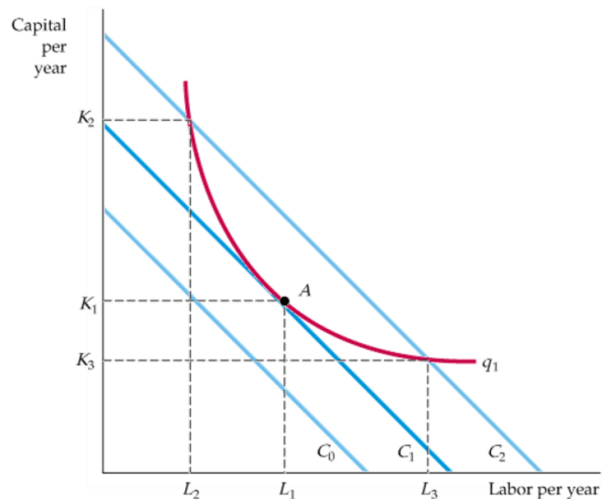
It follows that the isocost line has a slope of $\Delta K/\Delta L = -(w/r)$, which is the ratio of the wage rate to the rental cost of capital.



Cost in the Long Run: Choosing inputs

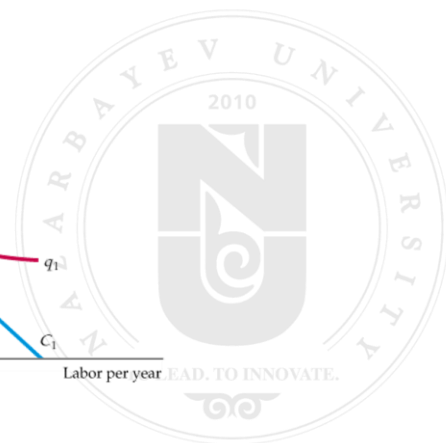
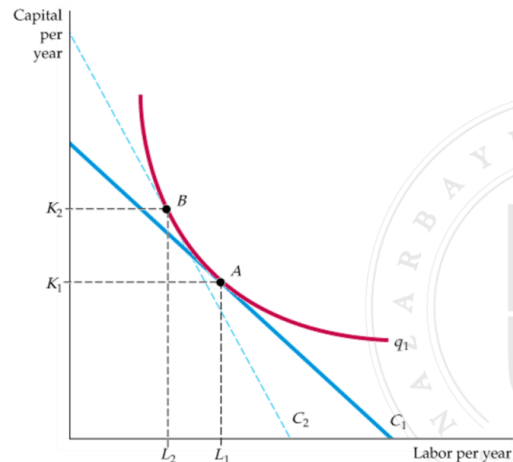
PRODUCING A GIVEN OUTPUT AT MINIMUM COST

Isocost curves describe the combination of inputs to production that cost the same amount to the firm. Isocost curve C_1 is tangent to isoquant q_1 at A and shows that output q_1 can be produced at minimum cost with labor input L_1 and capital input K_1 . Other input combinations— L_2, K_2 and L_3, K_3 —yield the same output but at higher cost.



INPUT SUBSTITUTION WHEN AN INPUT PRICE CHANGES

Facing an isocost curve C_1 , the firm produces output q_1 at point A using L_1 units of labor and K_1 units of capital. When the price of labor increases, the isocost curves become steeper. Output q_1 is now produced at point B on isocost curve C_2 by using L_2 units of labor and K_2 units of capital.



Cost in the Long Run

Recall that in our analysis of production technology, we showed that the marginal rate of technical substitution of labor for capital (MRTS) is the negative of the slope of the isoquant and is equal to the ratio of the marginal products of labor and capital:

$$\text{MRTS} = -\Delta K / \Delta L = \text{MP}_L / \text{MP}_K$$

It follows that when a firm minimizes the cost of producing a particular output, the following condition holds:

$$\text{MP}_L / \text{MP}_K = w / r$$

We can rewrite this condition slightly as follows:

$$\text{MP}_L / w = \text{MP}_K / r$$



Cost in the Long Run

Cost Minimization with Varying Output Levels

expansion path Curve passing through points of tangency between a firm's isocost lines and its isoquants.

To move from the expansion path to the cost curve, we follow three steps:

1. Choose an output level represented by an isoquant. Then find the point of tangency of that isoquant with an isocost line.
2. From the chosen isocost line, determine the minimum cost of producing the output level that has been selected.
3. Graph the output-cost combination in Figure 7.6 (b).

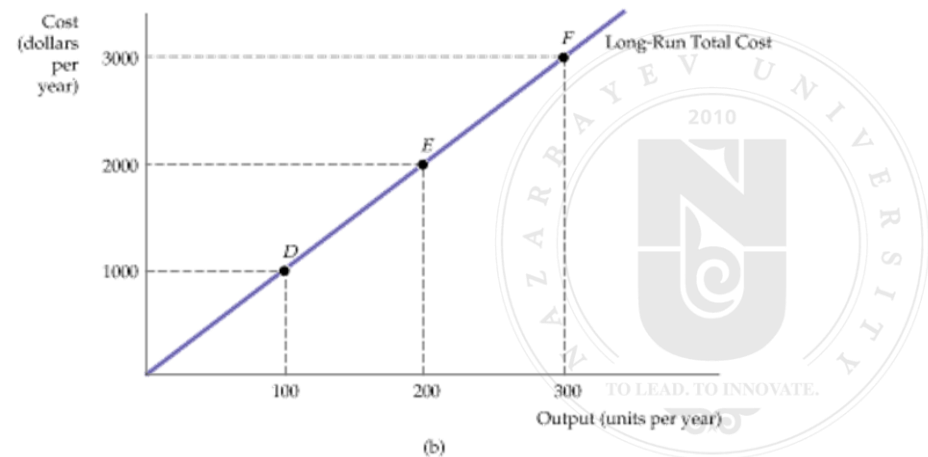
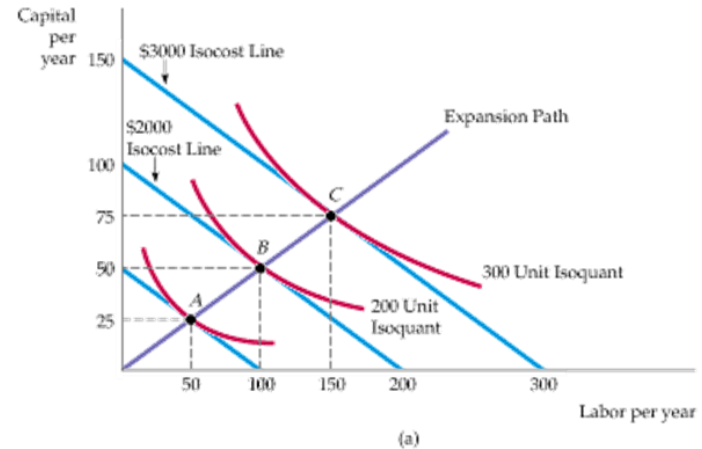


Cost in the Long Run

A FIRM'S EXPANSION PATH AND LONG-RUN TOTAL COST CURVE

In **(a)**, the expansion path (from the origin through points A, B, and C) illustrates the lowest-cost combinations of labor and capital that can be used to produce each level of output in the long run— i.e., when both inputs to production can be varied.

In **(b)**, the corresponding long-run total cost curve (from the origin through points D, E, and F) measures the least cost of producing each level of output.



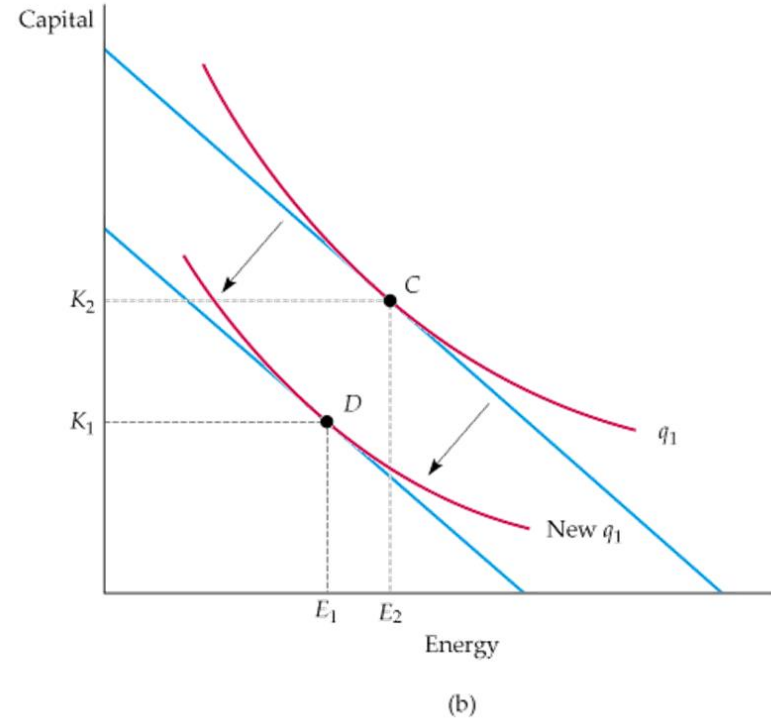
EXAMPLE

REDUCING THE USE OF ENERGY

ENERGY EFFICIENCY THROUGH TECHNOLOGICAL CHANGE

Technological change implies that the same output can be produced with smaller amounts of inputs. Here the isoquant labeled q_1 shows combinations of energy and capital that will yield output q_1 ; the tangency with the isocost line at point C occurs with energy and capital combinations E_2 and K_2 .

Because of technological change the isoquant shifts inward, so the same output q_1 can now be produced with less energy and capital, in this case at point D, with energy and capital combination E_1 and K_1 .



Long-Run versus Short-Run Cost Curves

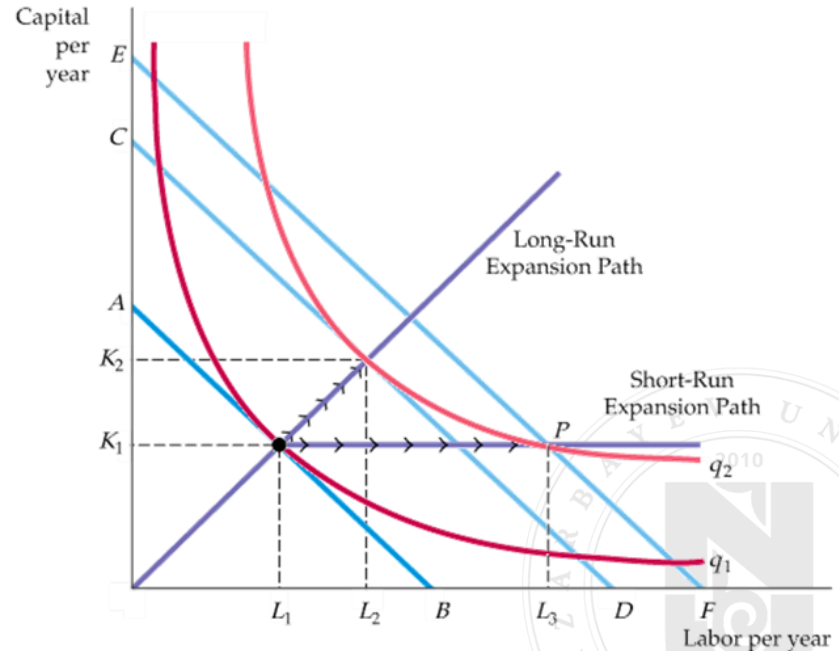
The Inflexibility of Short-Run Production

When a firm operates in the short run, its cost of production may not be minimized because of inflexibility in the use of capital inputs.

Output is initially at level q_1 , (using L_1, K_1).

In the short run, output q_2 can be produced only by increasing labor from L_1 to L_3 because capital is fixed at K_1 .

In the long run, the same output can be produced more cheaply by increasing labor from L_1 to L_2 and capital from K_1 to K_2 .



Long-Run versus Short-Run Cost Curves

Long-Run Average Cost

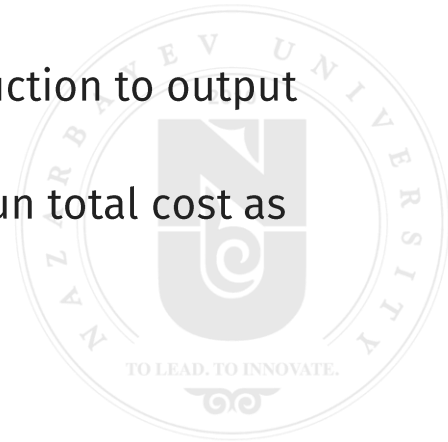
In the long run, the ability to change the amount of capital allows the firm to reduce costs.

The most important determinant of the shape of the long-run average and marginal cost curves is the relationship between the scale of the firm's operation and the inputs that are required to minimize its costs.

long-run average cost curve (LAC) Curve relating average cost of production to output when all inputs, including capital, are variable.

short-run average cost curve (SAC) Curve relating average cost of production to output when level of capital is fixed.

long-run marginal cost curve (LMC) Curve showing the change in long-run total cost as output is increased incrementally by 1 unit.



7.4 Long-Run versus Short-Run Cost Curves (3 of 8)

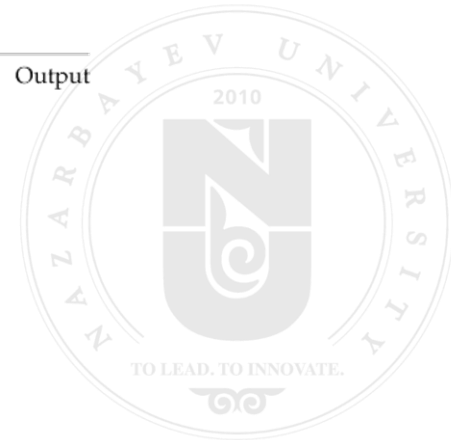
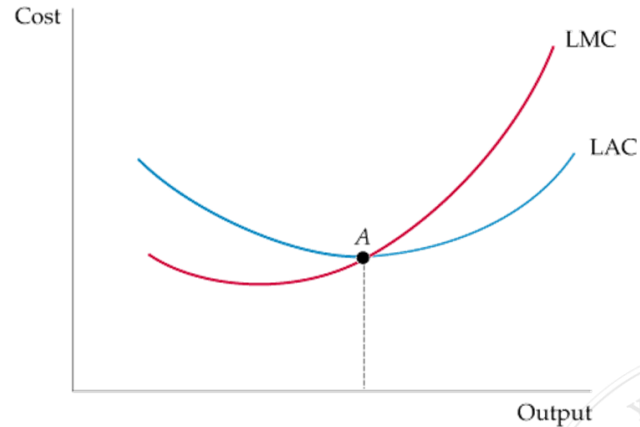
FIGURE 7.9

LONG-RUN AVERAGE AND MARGINAL COST

When a firm is producing at an output at which the long-run average cost LAC is falling, the long-run marginal cost LMC is less than LAC.

Conversely, when LAC is increasing, LMC is greater than LAC.

The two curves intersect at A, where the LAC curve achieves its minimum.



Long-Run versus Short-Run Cost Curves

Economies and Diseconomies of Scale

As output increases, the firm's average cost of producing that output is likely to decline, at least to a point.

This can happen for the following reasons:

1. If the firm operates on a larger scale, workers can specialize in the activities at which they are most productive.
2. Scale can provide flexibility. By varying the combination of inputs utilized to produce the firm's output, managers can organize the production process more effectively.
3. The firm may be able to acquire some production inputs at lower cost because it is buying them in large quantities and can therefore negotiate better prices. The mix of inputs might change with the scale of the firm's operation if managers take advantage of lower-cost inputs.

At some point, however, it is likely that the average cost of production will begin to increase with output.

There are three reasons for this shift:

1. At least in the short run, factory space and machinery may make it more difficult for workers to do their jobs effectively.
2. Managing a larger firm may become more complex and inefficient as the number of tasks increases.
3. The advantages of buying in bulk may have disappeared once certain quantities are reached. At some point, available supplies of key inputs may be limited, pushing their costs up.

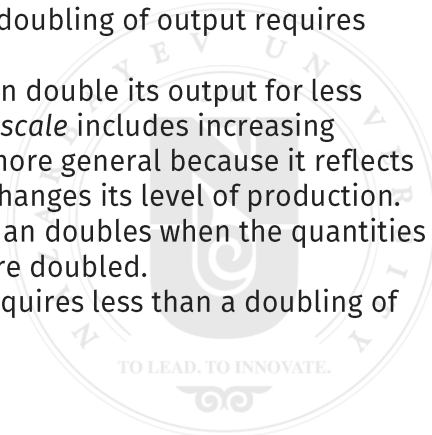
economies of scale Situation in which output can be doubled for less than a doubling of cost.

diseconomies of scale Situation in which a doubling of output requires more than a doubling of cost.

A firm enjoys **economies of scale** when it can double its output for less than twice the cost. The term *economies of scale* includes increasing returns to scale as a special case, but it is more general because it reflects input proportions that change as the firm changes its level of production.

Increasing Returns to Scale: Output more than doubles when the quantities of all input are doubled.

Economies of Scale: A doubling of output requires less than a doubling of cost.

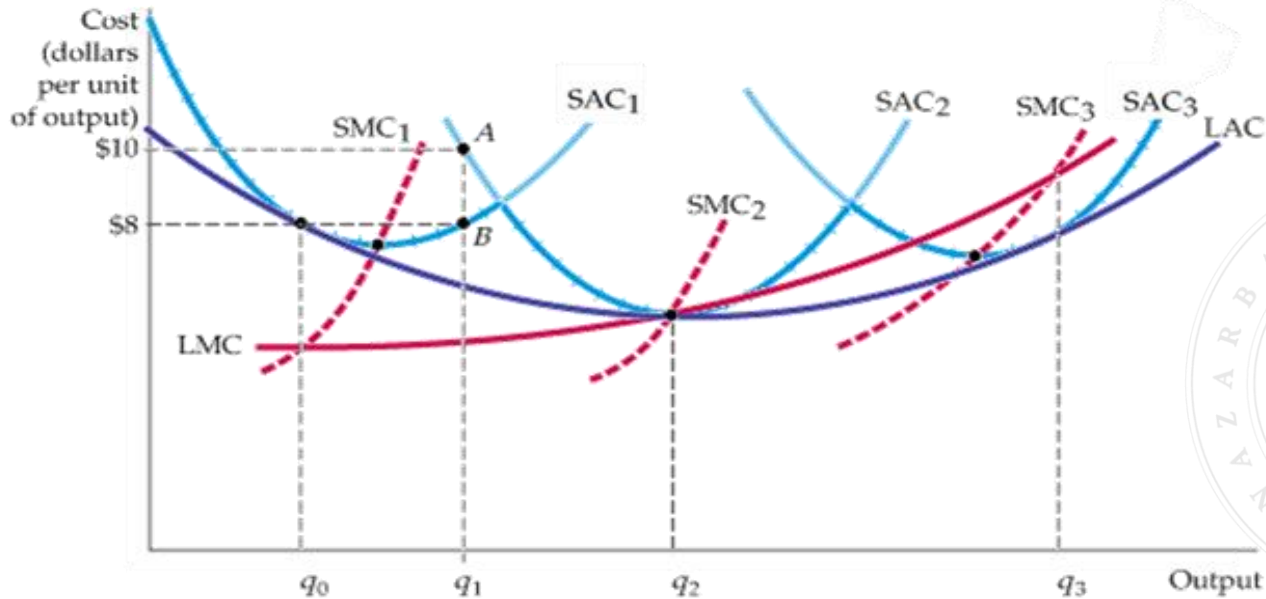


Long-Run versus Short-Run Cost Curves

LONG-RUN COST WITH ECONOMIES AND DISECONOMIES OF SCALE

The long-run average cost curve LAC is the envelope of the short-run average cost curves SAC_1 , SAC_2 , and SAC_3 .

With economies and diseconomies of scale, the minimum points of the short-run average cost curves do not lie on the long-run average cost curve, except SAC_2 which is the efficient level of output in the long-run





End of this Lecture.

Based on Chapters 6 & 7 of Pindyck and Rubinfeld.